First Demonstration of CVD Monolayer WSe_2 PFETs with Ultra-Short Channel Length of 16 nm: Achieving Record I_{on} of 1532 μA/μm at V_{ds} = -1 V and Ballistic Efficiency of 81%

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2026 IEEE/JSAP Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits) | 979-8-3315-8096-4/26/$31.00 ©2026 IEEE | DOI: 10.1109/VLSITECHNOLOGYANDCIR65830.2026.11577334

Abstract: In this work, we report an oxygen p-doping and full contact structure to reduce the contact resistance (R_c) of monolayer p-type tungsten diselenide (WSe_2) and demonstrate high-performance p-type WSe_2 field-effect transistors (FETs) with a scaled channel length (L_{ch}) of 16 nm. The controllable oxygen doping and the full contact structure greatly increase the contact quality. The co-optimized R_c is as low as 352 Ω·μm at 4.2 K, which is ~two times lower than that of the control sample with top contact. Consequently, 16-nm-channel WSe_2 pFETs deliver a record saturation current of 1532 μA/μm at a drain voltage of -1.0 V and a ballistic ratio of 81% at room temperature. This work presents a major breakthrough in p-type two-dimensional (2D) transistors in terms of contact optimization and device performance, filling the research gap in 2D p-type transistors.

Introduction: To realize 2D CMOS logic circuits, p-type 2D transistors with comparable performance to that of NFETs are essential. WSe_2 is a promising 2D p-type semiconductor due to its high intrinsic hole mobility and relatively large band gap. However, the selenium vacancy (V_{se}) in the WSe_2 film grown by CVD and the damage caused by metal contact formation can lead to undesired large Schottky barrier heights and R_c (Fig. 1a), which severely limits the electrical performance of WSe_2 PFETs, especially short-channel devices (Fig. 1b) [1,2]. Therefore, the on-state current (I_{on}) of monolayer WSe_2 PFETs is typically one order of magnitude smaller than that of MoS_2 NFETs (Fig. 1c) [3-11], which significantly limits the full utilization of 2D CMOS below the 1-nm technology node. In this work, we demonstrate ultralow R_c in monolayer WSe_2 PFETs through oxygen doping and a full-contact design (Fig. 2). The WSe_2 transistors achieve a record high saturation current (1532 μA/μm) and ballistic transport (81% ratio) at room temperature. The main achievements of this work are illustrated in Fig. 2.

Experiments: Monolayer WSe_2 was grown on 15 nm SiO_2 substrates by low-temperature chemical vapor deposition (LPCVD) in a quartz tube furnace. Molten-salt-assisted synthesis of WSe_2 was performed at 890 °C for 8 min in an Ar/H_2 atmosphere. The fabrication process flow of the monolayer WSe_2 PFETs is presented in Fig. 5a. The channel region was patterned by electron beam lithography (EBL) and isolated by reactive-ion-etching with a 100/10/80 sccm flow of $Ar/O_2/CF_4$ gas at 9.5 W power for 3 s. After annealing at 250 ℃ for 2 hours with 100 sccm Ar to remove AR-P residuals, S/D ohmic contacts were defined by EBL on the PMMA A4 resist and formed with Pt (10 nm)/Au (20 nm) deposited by high vacuum e-beam evaporation. After the metals from unwanted regions were lifted with acetone, the transistors were annealed at 200 ℃ for 2 hours in an O_2 atmosphere.

Results and discussion: For the top contact commonly used (Fig.3a), two current injection paths are along the metal edge and across the overlap. The full contact has an additional current injection path over the side contact—from the WSe_2-edge (Fig.3b). Figs. 3c and 3d show a simulated contour map of the current density integrated along the transport direction at the Fermi level in the top contact and full contact. The additional path of edge contact in full contact contributes to the overall current density which is greater than that of the top contact. Therefore, it is expected that full contact results in better contact quality with a low R_c and high current. The photoluminescence (PL) and Raman spectra depicted in Fig. 4 demonstrate the optical properties of monolayer WSe_2 and p-doping by oxygen annealing (O-doping), which is consistent with previous work [12,13]. Fig. 3b shows the I_d-V_g characteristics of 230-nm-channel monolayer WSe_2 transistors with and without O-doping at V_{ds} = -1 V. The transistors with O-doping clearly exhibit not only a higher drain current but also a lower subthreshold swing (SS). The I_{on} increases from 337 to 664 μA/μm and the SS decreases from 322 to 242 mV/dec after O-doping (Figs. 5c and 5d). This finding shows that oxygen annealing can simultaneously decrease the R_c and the interface trap density, which is crucial to high-performance short-channel transistors. To investigate the contact quality of two different structures, low-temperature measurements were performed. Figs. 6a and 6b show the I_d-V_g characteristics of WSe_2 transistors with different channel lengths (L_{ch}). We use the TLM structure to extract R_c as shown in Fig. 6c. R_c decreases with increasing carrier density (n_p) because of the increased carrier doping from the back-gate voltage (Fig. 6c). The minimal R_c of monolayer WSe_2 with full contact at 4.2 K is 352 Ω·μm at n_p = 1.6×10^{13} cm^{-2}, which is much smaller than the value for top contact WSe_2 (773 Ω·μm). To further evaluate the potential of monolayer WSe_2 PFETs, we scaled the channel length down to 16 nm, which was confirmed by high-resolution cross-sectional transmission electron microscopy (TEM) image (inset of Fig. 7a). Fig. 7a shows the transfer characteristic of ultrashort-channel monolayer WSe_2 transistor with 16-nm-long channel at room temperature, which has a high on/off ratio of ~1.8×10^6 at V_{ds} = -0.5 V. The corresponding output characteristics are shown in Fig. 7b, which reveals a record high saturation current of 1532 μA/μm at V_{ds} = -1.0 V. Fig. 8a illustrates the carrier transport in the ballistic and diffusive transistors. Ballistic transport becomes significant in ultrashort nanoscale transistors where the channel length is comparable to or shorter than the mean free path. Under these conditions, electrons travel through the channel without significant scattering events. Fig. 8b shows the modeling results of the output characteristic curves at room temperature with the standard ballistic Virtual Source Model (MIT), which agree well with the experimental data. The extracted ballistic efficiency of the fabricated 16-nm device is as high as 81%. Figs. 9a and 9b shows the low frequency noise of monolayer WSe_2 transistors with and without O-doping. The current noise spectral density (S_{id}/I_d^2) nicely follows a 1/*f* trend at all the gate voltages without the emergence of generation-recombination bulge signatures. To investigate the dominant physical mechanism of the *1/f* noise, S_{id}/I_d^2 and transconductance to drain current squared $(g_m/I_d)^2$ as functions of the drain current I_d of the same two devices are plotted in Figs. 9c and 9d, indicating that the main source of the 1/*f* noise of the WSe_2 transistor with O-doping is the carrier number fluctuation [14]. To investigate the reliability of O-doping, we performed high-temperature measurements from 25 to 125 ℃ and the results are shown in Fig. 10. High temperatures clearly do not cause obvious degradation, including I_{on} and off-state current. This is because low-temperature O_2 annealing effectively passivates selenium vacancies with oxygen instead of physical absorption, resulting in stable device performance. To make a fair comparison, we plot the drain current of monolayer 2D materials and conventional semiconductors as a function of V_{ds}/L_{ch}, which is essentially the average lateral electric field along the channel (Fig. 11a) [7,17-23]. We achieve the highest hole drain current at the same electric field. Fig. 11b shows the benchmark ballistic ratios of our WSe_2 PFETs with those of silicon MOSFETs, including 2D FETs, bulk Si FET, FinFETs, and double-gate (DG) Si FETs [21,24-26]. The ballistic ratio of our typical 16 nm WSe_2 PFET is significantly higher than those of all the previously reported silicon-based and other p-type 2D transistors.

Conclusion: In conclusion, our work resolves the critical challenge of low contact resistance in monolayer WSe_2 PFETs through oxygen doping and full-contact geometry. The monolayer WSe_2 transistor achieves a record high saturation current (1532 μA/μm) and ballistic transport (81% ratio) at room temperature. Our results prove that the expected p-type performance of WSe_2 is achievable for high-performance 2D p-type electronics for sub1-nm CMOS technology.

Acknowledgements: This work is supported by the National Key Research and Development Program of China (2024YFA1611200).

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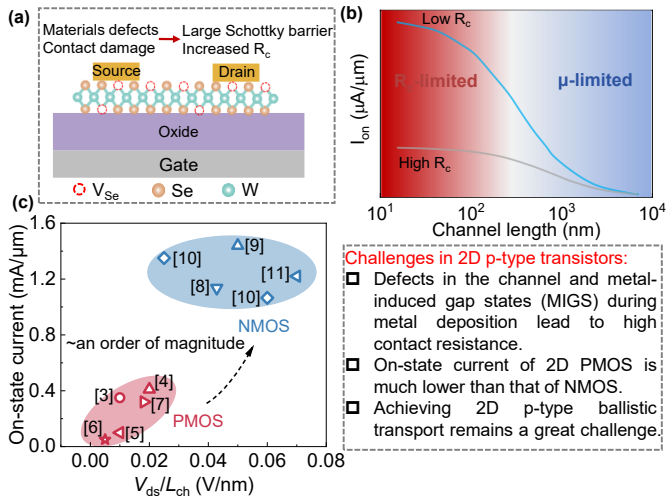
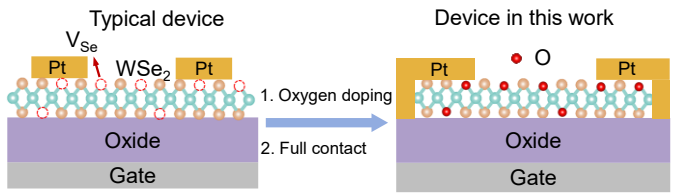


Fig. 1. (a) Illustration of materials quality and device fabrication process-induced defects in WSe_2 PFETs. (b) Schematic diagram of the effect of R_c on the drain current of transistors. (c) Comparison of the on-state current of typical MoS_2 NFETs and WSe_2 PFETs [3-11].



Highlights of this work:

- Ultralow contact resistance of 352 Ω·μm at 4.2 K is achieved in monolayer p-type WSe_2 by oxygen doping and full contact structure.
- Demonstration of a record high on-state current of 1523 μA/μm at a drain voltage (V_{ds}) of -1.0 V.
- First demonstration of CVD monolayer WSe_2 PFETs with a ballistic ratio of 81% at room temperature.

Fig. 2. Schematic diagram of the typical device structure used in the literature. Here, we use two methods to co-optimize the contact resistance of p-type WSe_2: (1) oxygen-doping; (2) full contact geometry. Summary of the key contributions of this work (the bottom panel).

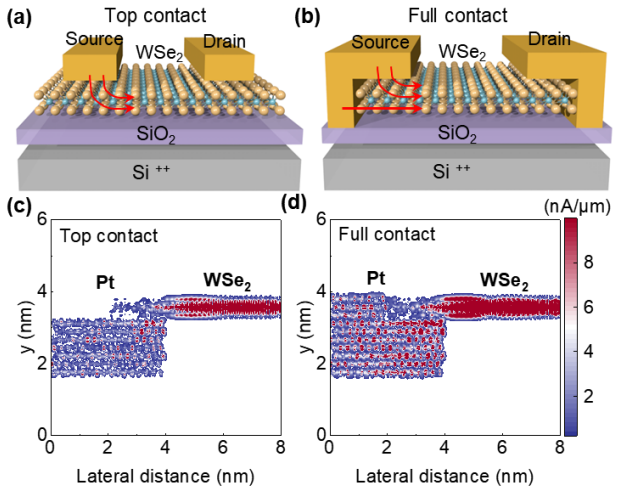


Fig. 3. Diagram of monolayer WSe_2 transistors with top contact (a) and full contact configurations (b). A simulated contour map of the current density in top contact (c) and full contact (d). The additional path of edge contact in full contact contributes to the overall current density, which is greater than that of the top contact.

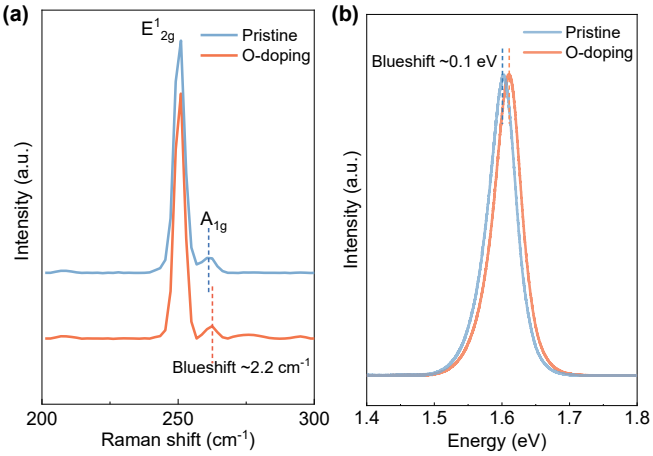


Fig. 4. (a) Raman and (b) PL spectra of monolayer WSe_2 with or without O_2 doping. The blueshift of the Raman A_{1g} peak and the PL peak after O-doping confirmed the p-type doping.

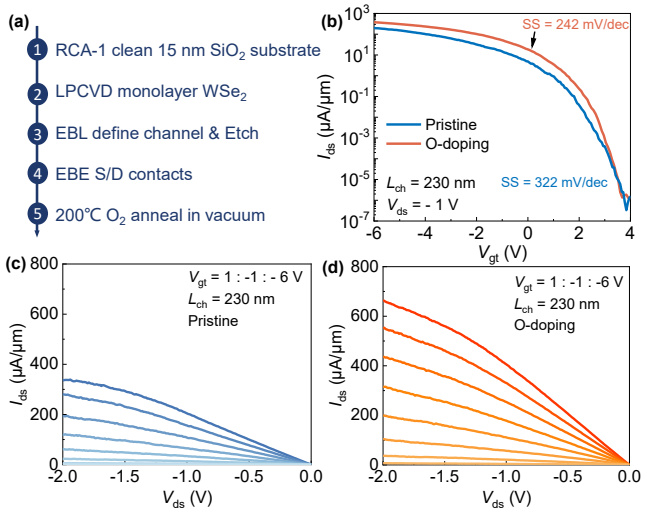


Fig. 5. (a) Fabrication process flow for p-type monolayer WSe_2 transistors. (b) Transfer characteristics of monolayer WSe_2 transistors with and without oxygen doping. (c), (d), Corresponding output characteristics for the same two devices in Fig. 5b.

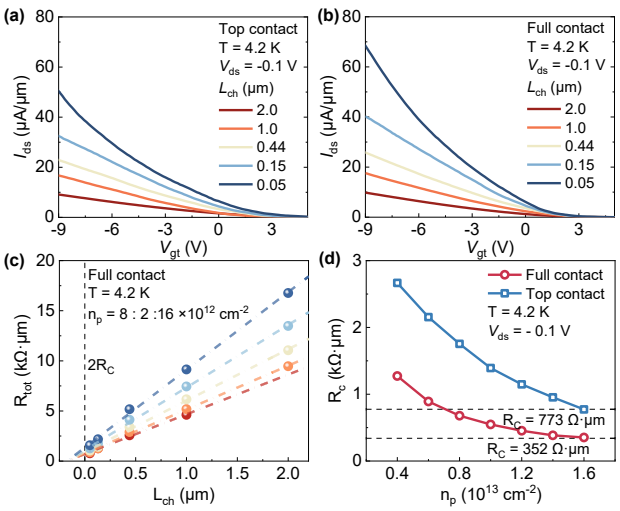


Fig. 6. Transfer curves of WSe_2 TLM devices with (a) top contact structure and (b) full contact structure. (c) Extraction of R_c by TLM (L_{ch}: 50/150/440/1000/2000 nm) for full contact WSe_2 PFETs. (d) R_c of two contact modes versus channel carrier concentration.

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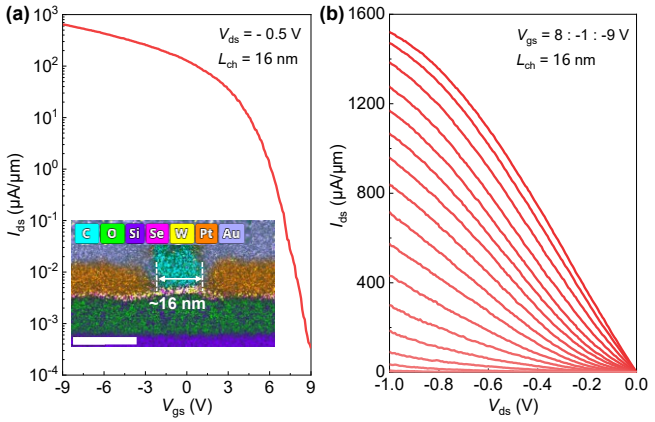


Fig. 7. (a) Room-temperature transfer characteristics of a typical 16-nm monolayer WSe_2 PFET on 15-nm-thick SiO_2 dielectrics. The inset shows EDS mapping of the device, scale bar: 20 nm. (b) Corresponding saturation output characteristics for the device in Fig.7a.

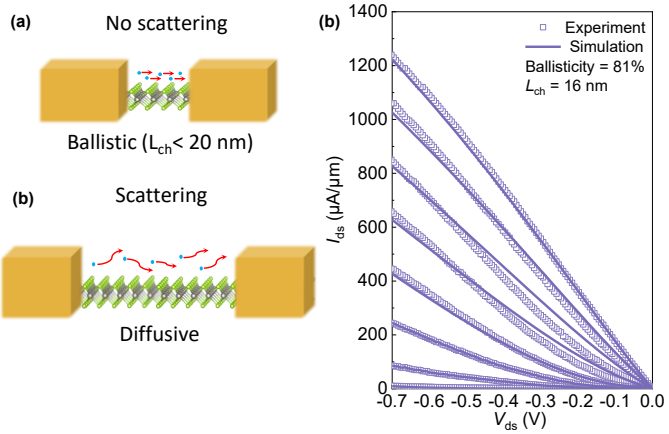


Fig. 8. (a) Ballistic transport becomes significant in ultrashort nanoscale transistors where the channel length is comparable to or shorter than the mean free path. Diffusive transport occurs when the transistor channel length is significantly longer than the mean free path of holes. (b) Comparison of the VS model fitting (lines) and measured data (dots) for output characteristics from a typical 16-nm-channel-length monolayer WSe_2 FET.

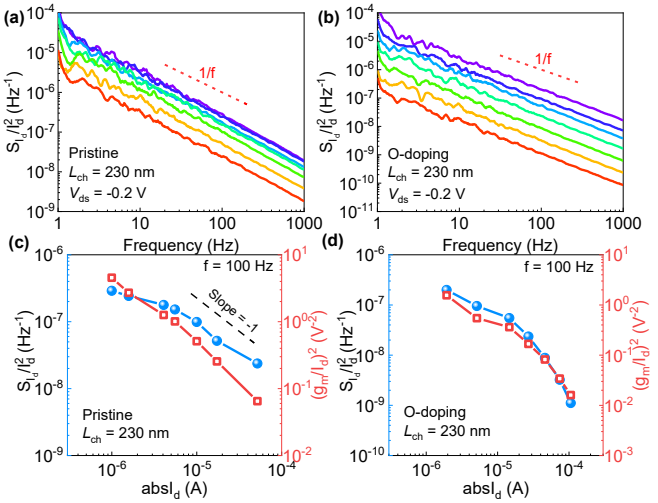


Fig. 9. S_{Id}/I_d^2 vs frequency of monolayer WSe_2 transistors with (a) without doping and (b) with O-doping. S_{Id}/I_d^2 and $(g_m/I_d)^2$ versus I_d for the WSe_2 device (c) without doping and (d) with O-doping. The noise mechanisms of WSe_2 transistors without and with doping are mobility fluctuation and carrier number fluctuation, respectively.

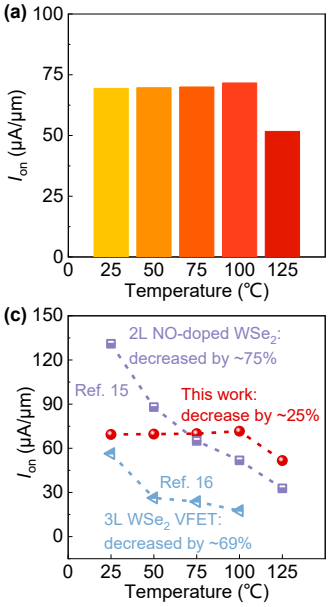
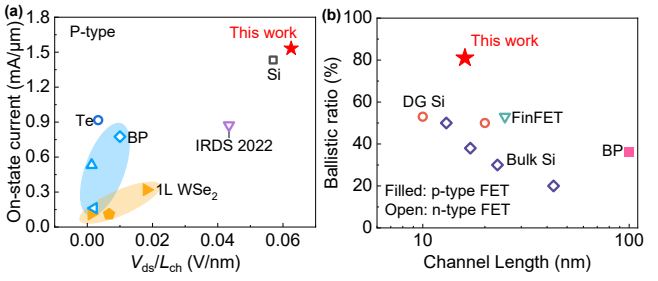


Fig. 10. On-state current (a), Off-state current (b), and subthreshold swing (SS) of a monolayer WSe_2 transistor measured at 25, 50, 75, 100 and 125 °C. The channel length is 1 μm. (d) Comparison of stability of our WSe_2 transistor with the others [15,16].



Channel	Contact	Dielectric	L_{ch} (μm)	I_{on} (μA/μm) V_{ds}=-1 V	Ref.
1L WSe_2	Pt	15 nm SiO_2	0.016	1523	This work
1L WSe_2	Ti/Pt	3.5 nm HfO_2	0.055	320	7
1L WSe_2	Ti/Pd	6 nm HfO_2	0.15	110	17
Si	-	-	0.014	1433 (V_{ds}=-0.8 V)	18
11.1 nm Te	Pd/Au	20 nm HZO/2 nm Al_2O_3	0.3	917	19
8.4 nm BP	Ni	90 nm SiO_2	1.5	300	20
12.5 nm BP	Ni/Au	HfLaO (EOT~2.7 nm)	0.1	775	21
11 nm BP	Ni/Au	20 nm HfO_2	0.5	162	22

Fig. 11. (a) The highest on-state current obtained from various p-type materials and devices [7,17-23]. (b) Benchmarking the room-temperature ballistic ratios of our WSe_2 PFETs with those of short-channel FETs, including bulk silicon FETs, silicon FinFETs, dual-gated silicon FETs and other 2D pFETs [21,24-26].

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